Topic: Introduction to Relational Algebra

Q. What is Relational Algebra in DBMS? Explain its basic concepts and categorize the primary relational algebra operators.

> 📌 **Definition to Remember**
> **Relational Algebra** is a **procedural query language** that forms the theoretical foundation of SQL. It specifies both *what* data is needed and *how* to retrieve it through a sequence of operations on relations (tables), where every operation takes one or two relations as input and produces a new relation as output.

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1. Basic Concepts

Concept	Explanation
Input/Output	Every operation takes 1 or 2 **relations (tables)** as input and produces a **new relation** as output.
Closure Property	Since output is always a relation, operations can be **nested and chained** in sequence.
No Side Effects	Operations only **query** data — they never modify, insert, or delete from the original tables.

2. Relational Algebra Operators

A. Basic (Fundamental) Operators

(All other operators can be derived from these)

Operator	Symbol	Description
Selection	σ	Extracts specific **rows** satisfying a condition
Projection	π	Extracts specific **columns**, discarding others
Union	∪	Combines rows from two relations, removes duplicates
Set Difference	-	Rows in first relation NOT in the second
Cartesian Product	×	Combines every row of A with every row of B
Rename	ρ	Renames a relation or its attributes

B. Derived Operators

(Combinations of basic operators for convenience)

Operator	Symbol	Description
Intersection	∩	Rows present in **both** relations (derived from -)
Join	⋈	Combines related tuples from two relations (Cartesian × Selection)
Division	÷	Answers "for all" queries (e.g., students who took **all** courses)

⭐ **Must-Write Points (for 10 marks)**

1. Relational Algebra is a procedural query language — the mathematical foundation of SQL.
2. Every operation takes 1 or 2 relations as input and always outputs a new relation (**Closure Property**).
3. Operations do not modify the original tables — they are read-only.
4. Basic operators: Selection (σ), Projection (π), Union (∪), Set Difference (−), Cartesian Product (×), Rename (ρ).
5. Derived operators: Intersection (∩), Join (⋈), Division (÷).
6. Derived operators can be expressed using combinations of basic operators.
7. Understanding Relational Algebra helps understand how the DBMS engine executes and optimizes SQL queries.

⚡ **Quick Recall**

`Procedural Language → Closure Property → Basic (σ, π, ∪, −, ×, ρ) → Derived (∩, ⋈, ÷) → Foundation of SQL`